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Abstract

The Philippine government implemented its conditional cash transfer program (CCT) in 2008, becoming its flagship strategy against poverty. The program achieves some of its short-term targeted outcomes primarily in consumption. This study focuses on non-targeted behavioral outcomes such as enterprise participation. This study seeks to determine if CCT has a positive impact on enterprise participation. Beneficiary households are obliged to spend based on the conditions of the program, but households are still confronted with different incentives on how to spend cash grants. Most CCTs have no specific rules about investing in businesses. After finishing the CCT program, entrepreneurial behaviors are important to maintain in order to improve their living standards. Using the 2014 Annual Poverty Indicator Survey and a quasi-experiment method, results reveal that CCTs have a positive impact on participation in business. The program increases the likelihood of household participation in enterprises and the number of enterprises they participated. Hence, policymakers can integrate a comprehensive mechanism into the CCT program to promote entrepreneurial capabilities.

Keywords: conditional cash transfer, business, enterprise, quasi-experiment, inverse probability weighting

Introduction

Conditional Cash Transfer (CCT) programs have been a preferred safety net against poverty in developing countries over the past decades (Shei, Costa, Reis, & Ko, 2014). Since its inception in Mexico in 1997, every country in Latin America now has its version of the program (Tutor, 2014). Approximately 30 countries have implemented a CCT program (Fiszbein & Schady, 2009). In the Philippines, over a quarter of the country's population is considered poor, and the poverty rate has never declined amidst modest economic growth (Chaudhury, Friedman, & Onishi, 2013). Thus, the Philippine government implemented its version of CCT in 2008 — the *Pantawid Pamilyang Pilipino Program* (4Ps). Since then, the program has expanded to include more sectors of the poor. In 2019, the program was institutionalized and has since become the primary anti-poverty measure of the government.

CCT programs provide cash to poor households on the condition that they invest in education and health (Fiszbein & Schady, 2009). The program targets the poorest segment of the population, particularly those households with school-aged children and pregnant mothers (Tutor, 2014). It aims to break the intergenerational cycle of poverty. Children of poor parents do not have enough access to education and health facilities and have higher chances of also becoming poor themselves (Shei et al., 2014). Lack of education and poor health are identified as the root cause of poverty (Canare, 2017). The program also aims to reduce the incidence of hunger and malnourishment among household members, which decreases their productivity and cognition (Tutor, 2014). Beneficiary households receive a maximum of Php 1,400 monthly, depending on the number of dependents and whether the mother is pregnant. This amount represents 13% to 26% of household expenses. The beneficiary household can receive a maximum of Php 15,000 in a year.

Empirical evidence suggests that the CCT programs achieve their targeted short-term outcomes. Consumption levels have risen significantly in some countries (Fiszbein & Schady, 2009). For instance, there is an observed 9% to 15% increase in per capita consumption in Paraguay (Soares, Ribas, & Hirata, 2010). Hoddinott and Skoufias (2004) report an increase in calories among beneficiaries than non-beneficiaries. Likewise, Attanasio and Mesnard (2006) find an increase in total monthly consumption by 15% in Colombia. Hagen-Zanker et al. (2016) observed an increase in food expenditure, school attendance, and utilization of health service facilities from several impact studies. Local impact studies suggest similar results. Chaudhury, Friedman, and Onishi (2013) find a strong positive impact of CCTs on school enrollment among young children, attendance to antenatal and postnatal care among poor pregnant women, and child health.

Studies suggest that cash transfers achieve other non-targeted outcomes, such as entrepreneurial behaviors. The economic activity of the beneficiaries may be the key to reduced dependence on the cash grant and maintaining their living standards after graduating from the program (Cirillo & Giovannetti, 2018). Gertler et al. (2012) find that households who invest part of their cash grant in productive activities increase long-term consumption.

Hence, this study aims to examine whether CCT achieves non-targeted outcomes. Specifically, it seeks to determine whether the CCT program positively impacts enterprise participation among poor households in the Philippines. Beneficiary households are obliged to spend based on the conditions of the program, but households are still confronted with different incentives on how to spend cash grants (Tutor, 2014). Most CCTs have no specific rules about investment in businesses (Ribas, 2019). After devoting a fraction of the cash grants to activities the program requires, the remaining amount can be used as collateral for borrowing or be saved for future use (Cirillo & Giovannetti, 2018). These behaviors, although important, are not the emphasis of most CCTs. Most CCTs focus on health and educational outcomes because poor health and lack of education are the main factors of intergenerational transmission of poverty (Chaudhury, Friedman, & Onishi, 2013).

In a review of several impact studies, Hagen-Zanker et al. (2016) notice that there are results indicating positive and negative effects on business participation. A negative impact means that CCT causes households to disinvest in business, likely increasing their dependence on cash transfers. Such behavior is an important issue among policymakers. If households become dependent on cash grants, there is the possibility of reverting back to preprogram levels of poverty once the beneficiaries graduate from the program (Gertler et al., 2012). Given the inconclusiveness of the relationship between CCT and this outcome, the topic remains an active area for research.

This study differs from previous studies, such as those of Canare (2017) and Tutor (2014), in three ways. First, this study focuses on non-targeted outcomes such as savings, borrowing, and business participation. Extending the analysis to include participation in enterprise and business gives another insight into the economic and financial outcomes of CCT. Earlier impact studies show that there is no noticeable increase in spending on more productive items, and financial inclusion does not improve (Chaudhury, Friedman, & Onishi, 2013). Second, this study evaluates CCT using a different data set and time frame. Lastly, this study uses a different impact evaluation methodology which is inverse probability weighting.

This study contributes to the small empirical literature on non-targeted outcomes. It provides empirical evidence about the causal effect of cash transfers on entrepreneurial behavior. Therefore, cash transfers can be used as a policy to stimulate the enterprise participation of households, especially among the poor. The development of these positive behaviors is vital in order to become self-sufficient and not reliant on

Methodology

Data and Sample

This study uses the 2014 APIS sourced from the PSA. It contains 1,243 variables, in which 10,469 households are interviewed. The objective of the APIS is to gather information on non-income indicators associated with poverty (Tutor, 2014). It contains information on socioeconomic variables such as demographic characteristics, housing conditions and tenure, source of utilities, sanitation facilities, assets, income, and expenditure (Tutor, 2014). In 2011, information on the social protection of households was included in the APIS, and one of the programs included is about the 4Ps. Most of the variables in the APIS can be used in the determination of the eligibility of the household into the cash transfer program (Tutor, 2014). Given the APIS information on household poverty indicators and 4Ps, the data has been used in several cash transfer studies in the Philippines. In particular, Tutor (2014) and Canare (2017) use the APIS 2011 to evaluate the impact of conditional cash transfers on consumption, savings, and borrowing behaviors. APIS also provides data on the productive activities of the households, such as their participation in the several entrepreneurial activities which this study also examines.

This study uses data from a sample of households. The variables used reflect the household behavior in general and not the individual members of the households. Financial behaviors such as savings, borrowing, and participation are at the household level. However, in cases where demographic variables are needed in the analysis, the demographic profile of the household head is used. The household head is the primary provider of the family. Participation in the program is based on household characteristics and demographic characteristics of the household head.

The distribution of households disaggregated by program participation is presented in Table 1.

Table 1. Distribution of beneficiary and non-beneficiary households across income brackets

Per Capita Income Decile	Beneficiary		Non-Beneficiary		Total
	Frequency	%	Frequency	%	
1	578	50.79	560	49.21	1,138
2	504	43.83	646	56.17	1,150
3	372	33.42	741	66.58	1,113
4	260	23.96	825	76.04	1,085
5	139	13.37	901	86.63	1,040
6	96	9.41	924	90.59	1,020
7	55	5.61	925	94.39	980
8	39	3.92	955	96.08	994
9	28	2.86	951	97.14	979
10	8	0.82	962	99.18	970
Total	2,079	19.86	8,390	80.14	10,469

Barrientos (2012) recommends focusing on the vulnerable group when assessing the impact of conditional cash transfers. The effect of cash transfers may be understated if the analysis includes all the samples. Thus, this study uses a subsample of households in the fourth per capita income decile and below. As much as possible, the sample must include only eligible households. Tutor (2014) and Canare (2017) observe the high likelihood of households belonging to the fourth income decile and below being selected for the program. The use of this subsample reduces the number of non-eligible households. There is a total of 4,486 households belonging to the fourth per capita income decile and below and 1,714 households (38.2%) who claim to be beneficiaries of the 4Ps.

Empirical Analysis

Inverse Probability Weighting

Inverse-probability weighting method uses the weighted average of the observed outcome variables to estimate the causal effect of the treatment using observational data. The weights used come from modeling the probability of treatment assignment from the subject characteristics using binary regression. This is called the treatment model. The treatment model should be correctly specified. The weight is the inverse of the probability of treatment assignment. The inverse probability weights correct for confounding and missing data problems. Observations that are likely to have no missing data get a weight close to one. On the other hand, observations containing missing data get a weight potentially larger than one. Individuals with a high probability of being exposed to the treatment have a lower weight than those with a low probability of exposure. Individuals with a low probability of being given a treatment but actually receiving a treatment represent a larger group of individuals who do not actually receive the treatment (Curtis et al., 2007). The inverse probability weighting creates a "pseudo-population" twice as large as the original population, where every individual appears to be both treated and untreated (Hernan & Robins, 2006). Thus, the method enables researchers to compare outcomes from the treated and untreated from this hypothetical population.

To put this formally, the unweighted unbiased estimator of the causal effect of the treatment is

$$\theta = \frac{1}{n_1} \sum_i^{n_1} y_{1i} - \frac{1}{n_0} \sum_i^{n_0} y_{0i} \quad (2)$$

where the subscript 1 and 0 indicates treatment assignment (1 if treated, 0 if untreated), n_1 is the number of individuals in the treated group, n_0 is the number of individuals in the untreated group, y_{1i} is the outcome for individual i for the treated group and y_{0i} is the outcome for individual i in the untreated group. The estimator θ is the average difference in the outcomes between the treated and control group. Confounding in observational data violates the conditional independence assumption (CIA) and matching

assumption resulting in a biased estimator θ^1 . Implementing IPW to control for confounding involves weighting the outcome for those treated by

$$w(x) = \frac{1}{p(x)} \quad (3)$$

and for those controlled (untreated) by

$$w(x) = \frac{1}{1-p(x)} \quad (4)$$

where w is the weight and $p(x)$ is the propensity score or conditional probability that the individual will be exposed to the treatment given covariates $\mathbf{x}$ denoted as $P(T = 1|X = x)$. The propensity score is estimated using logistic regression or probit regression model. Combining the weights and estimator θ gives the IPW estimator

$$\theta = \frac{1}{n_1} \sum_i^{n_1} \frac{y_{1i}}{p(x_i)} - \frac{1}{n_0} \sum_i^{n_0} \frac{y_{0i}}{1-p(x_i)} \quad (5)$$

In the calculations of the weights using binary models, it is possible that some of the probabilities are close to zero. Since the IPW uses the inverse of the propensity scores, the weight becomes arbitrarily large as the probability goes to zero. Individuals with a very low probability of being treated but who end up receiving the treatment have a very large weight. This makes the IPW estimator heavily dependent on very few samples. Thus, the IPW estimates become unstable. Most often, this increases the variance and makes the estimate unstable (Thoemmes & Ong, 2015). Researchers recommend the use of stabilized weights to solve this problem (Hernan, Brumback, & Robins, 2002). The key difference between the stabilized weights and the unstabilized ones is that the numerator is replaced with the proportion of individuals in the treated and control groups (Thoemmes & Ong, 2015). Therefore, the stabilized weight for those who are treated is

$$w_s(x) = \frac{P(T=1)}{p(x)} \quad (6)$$

and for the control is

$$w_s(x) = \frac{1-P(T=1)}{1-p(x)} \quad (7)$$

Stabilized weights preserve the sample size and produce estimates with smaller variances (Thoemmes & Ong, 2015). The IPW estimator with stabilized weight is

$$\theta = \frac{1}{n_1} \sum_i^{n_1} \frac{(P(T = 1))y_{1i}}{p(x_i)} - \frac{1}{n_0} \sum_i^{n_0} \frac{(1 - P(T = 1))y_{0i}}{1 - p(x_i)} \quad (8)$$

Usually, implementing an IPW involves a two-step process: (1) estimation of the propensity score and computation of the inverse probability weights; and (2) use of the estimated inverse probability weights to compute the weighted average of the outcomes for each treatment. The contrast of the weighted average outcomes is the average treatment effect.

Doubly Robust IPW

When the specification of the treatment model is not correct, the IPW estimator becomes less consistent. In this case, the double robust IPW estimators can be used. These estimators are the augmented IPW (AIPW) and IPW regression adjustment (IPWRA). These are called double robust estimators because these estimators include both the treatment and the outcome models. Only one of these models needs to be correctly specified for the estimator to be consistent.

The AIPW estimators combine both the properties of the regression adjustment estimator and the IPW estimator. It contains information about the probability of treatment assignment and the predictive model of the outcome. The AIPW estimator contains an augmentation term that corrects the estimator when the treatment model is misspecified. However, if the treatment model is correctly specified, the augmentation

¹ CIA state that the outcomes are independent of treatment. Matching assumption states that for every individual in the program there is a matched individual not selected in the program.

term disappears when the sample size becomes large. The augmentation term is the weighted average of the outcome model. The AIPW involves the following steps: (1) estimation of the parameters of the treatment model and computation of the inverse probability weights; (2) estimation of the regression models for the outcome for each treatment and obtaining treatment-specific predicted outcomes for each subject; (3) computation of the mean of the treatment-specific predicted outcomes weighted by the inverse probability from step 1. The contrast of the weighted average is the estimate of the average treatment effect.

In the IPWRA, the estimator uses a weighted regression coefficient to determine the average of treatment-specific predicted outcomes. The weights are the estimated inverse probability weights. The model in determining the treatment and outcome may be different. The steps involved in implementing an IPWRA are (1) estimation of parameters of the treatment model and computation of the inverse probability weights; (2) use of the inverse probability weights to fit a weighted regression model of outcomes for each treatment level and obtain the treatment specific predicted outcomes for each subject; (3) and computation of the means of the treatment-specific outcome. The contrast of these outcomes is the estimate of the average treatment effect. The IPWRA produces a consistent estimate of the treatment effect parameter because treatment is assumed to be independent of the potential outcome after conditioning on the covariates.

These estimators can accommodate whether the outcome is continuous, binary, or count variable. The outcome model can be specified to use OLS, logistic, or Poisson model depending on the type of the outcome variable.

Estimation of the Propensity Score

IPW estimators involve first the estimation of the propensity score—the probability that the household is selected to the 4Ps. This is the treatment model. The estimated conditional probability is used in the weighting of the outcomes. The dependent variable is the exposure to the conditional cash transfer program, which is binary—being selected or not selected to the CCT program. Therefore, a logistic regression model of the form

$$P(D = 1|X_1, X_2, \dots, X_k) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k)}} \quad (9)$$

is performed where the dependent variable D is the treatment indicator which is 1 if the household is a 4Ps beneficiary and 0 otherwise. The covariates X are household characteristics. This regression model is appropriate since this limits the dependent variable to 1 and 0. The logistic regression model has the following specification:

$$\begin{aligned} P(D = 1|X) = F(\beta_0 + \beta_1 \text{Male} + \beta_2 \text{Age} + \beta_3 \text{Married} + \beta_4 \text{Educ} \\ + \beta_5 \text{householdmember} + \beta_6 \text{Househeadwork} \\ + \beta_7 \text{spousework} + \beta_8 \text{roofing} + \beta_9 \text{wall} \\ + \beta_{10} \text{elect} + \beta_{11} \text{water} + \beta_{12} \text{Toilet} \\ + \beta_{13} \text{ownhouse} + \beta_{14} \text{rent} + \beta_{15} \text{floorarea} \\ + \beta_{16} \text{typehouse} + \beta_{17} \text{OFW} + \beta_{18} \text{NCR} \\ + \beta_{20} \text{appliance}) \end{aligned} \quad (10)$$

Selection of the Covariates

While the logistic regression conveniently solves for the conditional probability, the selection of the covariates is not. Choosing variables to estimate the propensity score is an iterative process and a “tricky” exercise because the covariates must be balanced between the control and treatment groups (Canare, 2017). It means that, on average, the treatment and control are the same on characteristics X . Often, this is the challenge most researchers face in treatment effect analysis. This process will determine if the CIA and matching assumption will hold.

Researchers suggest several ways to select the variables to include in the estimation of the propensity score, although there is no standard method in the current literature. Caliendo and Koepinig (2008) recommend using variables that are fixed over time or those measured before treatment assignment. They also suggest using variables that are informed by economic theory, previous empirical studies, and knowledge about the program. In the case of CCT, targeted households are regarded as the “poorest of the poor,” which is determined using a proxy mean test. Instead of using income, other indicators which are

easier to measure are used as a proxy to measure income. Literature suggests the following as close correlates of income: (1) needs indicator (e.g., household size, household composition, and dependency ratio); (2) measure of current command over resources (e.g., assets such as TV, appliances), housing amenities, marital status, and OFW member; (3) measure of potential command over resources (e.g., employment status, education); transfer from people, organization or government (e.g., public amenities); (4) economic opportunity (e.g., bank account, business enterprise); and (5) geographic location (e.g., urban and rural) (Mapa & Albis, 2013). Most of these variables have some representation in the APIS 2014.

The number of variables to be included is another issue to be decided by the researcher. Bryson, Dorsett, and Purdon (2002) discourage the use of too many variables. It may worsen the common support problem and increases the variance of the estimates, although it may not be biased. This problem is made worse if the sample size is small (Augurzyk & Schmidt, 2001). But the small sample size is not a problem when using a national household survey, and the chance of achieving balance in the covariates is high (Canare, 2017).

Balance Test

Rubin (2008) suggests checking the balance of the covariates first before looking into the estimated treatment effect parameter. The balance of the selected covariates is tested by comparing the means and examining the standardized differences and the variance ratio. A perfectly balanced covariate has a standardized difference of 0 and a variance ratio of 1. Imai and Ratkovic (2014) have developed a balance test that views the restrictions imposed by balance as an overidentifying condition that can be conveniently executed in statistical software. If covariates are not balanced, researchers suggest to consider doing the following and rerun the balance test: (1) add interaction and quadratic terms; (2) drop one or two covariates that are not theoretically important; (3) re-categorize variables; and (4) include higher order terms or spline of variables (Beal & Kupzyk, 2014).

Estimation of the Treatment Effect Parameters

In treatment effect analysis, there are three parameters of interest. These are the potential outcome means (POM), average treatment effect (ATE), and average treatment effect on the treated (ATET).

The POM is the average potential outcome for each treatment level. It can be denoted as $E(y_t)$, where t is the treatment level, with 1 indicating treatment and 0 no treatment. The POM for those who are treated is $E(y_1)$, and for those untreated is $E(y_0)$. In this study, potential outcome means are the average of the savings, borrowing, and productive activities outcomes.

The ATE is the average difference in the potential outcomes of the treated and the untreated. It is the average effect of the treatment on the population. In this study, for instance, it is the average difference in the savings, borrowing, and productive activities between households who are in the conditional cash transfer program and those who are not. If y_1 is the outcome of the treated group and y_0 is the outcome of the untreated group, then the ATE can be represented as $E[y_1 - y_0]$.

An alternative parameter usually computed by researchers is the average treatment effect on the treated (ATET). The ATET is the difference between the outcome of the program participants and the outcome of program participants if they had been excluded from the program, denoted as $E[y_1 | D = 1] - E[y_0 | D = 1]$ (W. Greene, 2012). It can be noted that $E[y_0 | D = 1]$ is also an unobservable component but can be estimated with additional assumptions (Tutor, 2014). For ATET to be valid, treatment assignment D must be uncorrelated with y_0 conditional on the propensity score $p(x)$ (Tutor, 2014). This implies that before participating in the program intervention, households eligible for the program must have the same expectation of the outcome as if they had not participated in the program. The outcome must not influence their participation. This assumption can be expressed as $E[y_0 | \text{Pr}(x), D = 1] = E[y_0 | \text{Pr}(x), D = 0]$, which is a weaker version of the CIA. The second additional assumption is a simplified version of the common support assumption, which is $\text{Pr}[D = 1 | x] < 1$. The condition $0 < \text{Pr}[D = 1 | \text{Pr}(x)]$ does not necessarily need to be satisfied in ATET because the interest is differences in outcome among those who participated in the treatment. ATET can be computed in matching, regression adjustment, and inverse probability weighting approaches. Advances in statistical computing allow the computation of ATET whether the outcome variable is continuous, categorical, and count in nature.

Checking for Sensitivity

Aside from using IPW, nearest-neighbor matching (NNM) is done to also estimate the impact of CCT on savings, borrowing, and participation in enterprises. Using different methods in treatment effect evaluation is a form of sensitivity analysis (Canare, 2017; Veras Soares et al., 2010). In NNM, each treated observation

is matched with the closest untreated comparison group in terms of the propensity score. If the ATET is significantly not the same for IPW and NNM, then it is not considered robust (Veras Soares et al., 2010).

Variable Definition

Table 2 summarizes the definition and measurement of variables used in the study.

Table 2 Variable definition and measurement

Variables	Definition	Measurement
Treatment	Selection of the household to the 4Ps program	1 if the household is a beneficiary of 4Ps, 0 otherwise
Outcome Variables		
Enterprise participation	Indicates whether the household participates in at least one business of any kind	1 if the household is engaged in any kind of business/enterprise, 0 otherwise
Number of enterprises	Reflects the household's inclination to engage in business	The actual number of any enterprise or business participated in by the household
Covariates		
Male	A demographic characteristic reflecting physiological distinction at birth	1 if male, 0 otherwise
Married	A demographic characteristic indicating the marital status of the household head	1 if married, 0 otherwise
Age	A demographic characteristic that reflects what stage of life the household head is	Number. of years living since birth
Elementary Level	A demographic characteristic that reflects the level of human capital	1 if the household head's educational attainment is at the most elementary level, 0 otherwise
Household head with work	Reflects the employment status of the household head	1 if the household head has work, 0 otherwise
Spouse with work	Reflects the employment status of the spouse	1 if the spouse has work, 0 otherwise
With household members ages 0 to 5 years old	Household composition	1 if the household has members ages 0 to 5 years old
With household members ages 6 to 14 years old	Household composition	1 if the household has members ages 6 to 14 years old
With household members ages 15 to 18 years old	Household composition	1 if the household has members ages 15 to 18 years old
With OFW member	Household composition	1 if the household has members who work abroad, 0 otherwise
Floor area	A housing characteristic that reflects housing and living conditions	The floor area of the housing unit in terms of square meters (sq.m.)
Roof made of light materials	A housing characteristic that reflects housing and living conditions	1 if the roofing of the house is made of light materials, 0 otherwise
Wall made of light materials	A housing characteristic that reflects housing and living conditions	1 if the walls of the house are made of light materials, 0 otherwise
Access to Electricity	A housing characteristic that reflects access to basic utilities	1 if the house has electricity, 0 otherwise

Variables	Definition	Measurement
Distance from the water source	A housing characteristic that reflects access to basic utilities	Distance of water source from the housing (m)
Household has no toilet facility	A housing characteristic that reflects access to the basic sanitary facility	1 if the household has no toilet facility, 0 otherwise
Household owns house and lot	A housing characteristic indicating ownership of the house and land	1 if the household owns the house and lot, 0 otherwise
Household paying rent	A housing characteristic indicating ownership of the house and land	1 if the household rents the house and lot, 0 otherwise
Single Type of House	A housing characteristic indicating housing and living conditions	1 if the type of building of the house is single, 0 if duplex, apartment, and other types of housing
Household resides in National Capital Region (NCR)	A housing characteristic reflecting the location of the residence	1 if the household resides in NCR, 0 otherwise
Number of Appliances	A household characteristic reflecting the living condition	The sum of all indicators for the 12 categories of appliances

The variable *Treatment* is a binary variable with a value of 1 if the household is a beneficiary of the 4Ps and 0 otherwise. The APIS 2014 solicits information about the CCT participation of each household member. It provides information on how many household members receive benefits from the program and how much they receive. This study considers a household as a beneficiary of the program if at least one member is a participant in the program.

For enterprise participation, a binary variable and count variable is also used, namely, *Enterprise participation* and *Number of enterprises*, respectively. A household is considered participating in an enterprise if there are earnings (negative or positive) in any of the following activities: crop farming and gardening; livestock and poultry raising; fishing; forestry and hunting; wholesale and retail; manufacturing; community, social, recreational, and personal services; transportation, storage, and communication services; mining and quarrying; construction; and other entrepreneurial activities not elsewhere classified. Although negative income denotes the poor performance of the enterprise, this, however, does not nullify the fact that the household engaged in a business enterprise.

A total of 21 variables are considered in calculating the propensity score. These variables are selected based on knowledge of the program and existing literature. These variables are considered strong correlates of income and strongly influence participation in the program. The variables are broadly categorized as demographic characteristics of the household head, household composition, employment, housing characteristics, basic utilities, asset ownership, and location. Almost all measures are dummy and indicator variables except for *Age*, *Floor area*, *Distance from water source*, and *Appliance*. The variable *Appliance* is the sum of all indicators for the 12 categories of appliances. Each category of appliances is coded as 1 if the household has this appliance and 0 otherwise. These appliances are an air con, washing machine, stove and gas range, refrigerator, personal computer, cellphone, telephone, stereo and audio component, videoke/karaoke, magic sing, DVD/VSD player, TV, and radio cassette player. The sum of all these indicators of appliances is calculated. The number indicates the total number of appliances owned by the households.

Results and Discussions

Descriptive Results

Household profile disaggregated by program participation

Table 3 summarizes the profile variables disaggregated according to program participation.

Table 3 Profile disaggregated by program participation

	Treatment Mean	
	Beneficiary (1,714 households)	Non-Beneficiary (2,772 households)
Demographic Profile		
Male	0.89	0.81
Married	0.88	0.77
Age	45.74	51.49
At most elementary level	0.63	0.57
With household members ages 0 to 5 years old	0.52	0.44
With household members ages 6 to 14 years old	0.94	0.56
With household members ages 15 to 18 years old	0.49	0.30
With OFW member	0.06	0.09
Household head with work	0.94	0.84
Spouse with work	0.53	0.41
Housing Characteristics		
Floor area (sq.m.)	34.83	37.57
Roofing made of light materials	0.25	0.16
Wall of house made of light materials	0.37	0.25
Water distance (m)	54.03	36.83
Household has no toilet facility	0.06	0.05
Household owns house and lot	0.60	0.60
Household paying rent	0.01	0.04
Single type of house	0.99	0.96
Household reside in NCR	0.02	0.04
Appliance	1.94	2.26
Entrepreneurial Behavior		
Households with participation in any enterprise	0.69	0.59
Number of enterprises of the households	0.92	0.76

Of the 4,486 households in the sample, only 38%, or 1,714 households, claimed to be beneficiaries of the 4Ps. The table reveals that the proportion of male (0.89%) and married (0.88) household heads is higher among beneficiary households than in non-beneficiary households. Moreover, heads of beneficiary households are younger (45.74 years old) than heads of non-beneficiary households (51.49 years old). The program targets households with members considered mothers; therefore, household heads are expected to be married, and the couples are in their fertile years. Most household heads among beneficiary households have attained the elementary level (0.63). Education is a significant predictor of being selected for the program. Low education is most often associated with low earnings and income. In terms of household composition, the proportion of households with members ages 0 to 5 (0.52%), 6 to 14 (0.94), and 15 to 18 (0.49) years old are higher for beneficiary households. This observation is expected since the CCT targets households with children ages 0 to 18 years old who are still dependent on their parents. The proportion of household heads with work is higher (0.94) for beneficiary households than non-beneficiary households (0.84%). It seems contradictory because heads of household beneficiaries are expected to be unemployed, contributing to their chances of being poor. A higher proportion (0.53) of household heads whose spouses work is observed for beneficiary households. These observations about the beneficiary household heads and working spouses are contrary to the belief that cash transfers will make household members work less (Chaudhury, Friedman, & Onishi, 2013).

As to housing characteristics, it is anticipated that household beneficiaries have poor housing conditions and lack access to some basic utilities. Results show that household beneficiaries have, on average, a smaller housing floor area (34.83 sq.m.), and a higher proportion have houses with roofs (0.25) and walls (0.37) that are made of light materials. Houses of almost all (0.99) of the beneficiaries are of a

single type. Housing tenure may not be an issue among the household beneficiaries. Sixty percent of household beneficiaries and non-beneficiaries own their lots and houses. A lower proportion of beneficiary households (1%) pays rent for houses and lots than the non-beneficiary households (0.04). Although this seems contradictory, this makes sense. Beneficiary households cannot afford to pay rent for houses and lots and settle in locations free from these types of expenses. Water is difficult for the household beneficiaries, as indicated by the long distance (54.03 m) to its source. More households among the beneficiaries (.06%) do not even have toilet facilities compared to non-beneficiaries (.05).

The mean of the other variables is also in accordance with the targeted households of the CCT program. Remittances from OFWs also contribute to the household not being poor. More households among non-beneficiaries (0.09) have OFW members than beneficiary households (0.06). Cash transfers from abroad are significant sources of funds for consumption. The APIS 2014 does not have indicators on the type of location, such as urban or rural. Instead, this study uses NCR as a proxy since it is highly urbanized. Results show a higher proportion of non-beneficiary households (0.04) located in NCR than among beneficiary households. Households in urban areas have higher chances of being non-poor because of significant economic activities. Beneficiary households are observed to own a lower number of household appliances (1.94) than non-beneficiary households. Non-beneficiary households can afford to buy more household appliances. In general, the profile disaggregated by treatment reveals the state of the targeted household of the CCT program. They live in houses with poor construction materials, lack access to basic utilities, do not have enough appliances, are without OFW members, and are located in rural areas.

Meanwhile, the result for entrepreneurial activities closely follows the proposition of this study. A higher proportion of beneficiary households (0.69) participate in some form of entrepreneurial activities compared to non-beneficiary households (0.59). Furthermore, beneficiary households participate in more enterprises (0.92) than compared to non-beneficiary households (0.75). These preliminary results imply a positive impact of CCTs on the financial profile of the beneficiary households.

Estimation of the Propensity Score

Table 4 shows the result of the estimation of the propensity score.

Table 4. Propensity score model estimate

Covariates	Dependent variable: Treatment=1 if the household is beneficiary, 0 otherwise	
	dy/dx	Std. Err.
Male	-0.026	0.027
Married	0.044	0.028
Age	0.008	0.004**
Agesquare	0.000	0.000***
Elementary	0.064	0.014**
Hmembers0_5dum	0.051	0.013***
Hmembers6_14dum	0.411	0.017***
Hmembers15_18dum	0.130	0.013***
Hhwork	0.013	0.026
Spousewrkdum	0.033	0.014**
Floor_area	0.000	0.000
Rooflight	0.063	0.018***
Walllight	0.042	0.016***
Water_distance	0.000	0.000**
Notoilet	-0.010	0.029
Ownhouselot	0.007	0.013
Renthlot	-0.174	0.045***
Singhousedum	0.159	0.051***
OFW	-0.050	0.025**
NCR	-0.075	0.047

Covariates	Dependent variable: Treatment=1 if the household is beneficiary, 0 otherwise	
	dy/dx	Std. Err.
Appliance	-0.025	0.004***

* p<0.1; ** p<0.05; *** p<0.01

This is also known as the treatment model estimation because the dependent variable is program participation. Since the propensity score is the conditional probability of being selected in the program given the covariates, it is convenient to report the average marginal effect. The average marginal effect in the context of logistic regression is the average change in the probability of the outcome for a unit change in the covariates. This reveals how the propensity score changes if there is a change in the covariates. In this case, the dependent variable is the household being a beneficiary of the 4Ps. Thus, the marginal effect is the average change in the probability of the household being selected to the 4Ps for every unit change in the covariates.

Results reveal positive significant signs for *Hmembers0_5dum* (households with members ages 0 to 5), *Hmembers6_14dum* (households with members aged 6 to 14), and *Hmembers15_18dum* (households with members aged 15 to 18). These variables represent household composition that has an important effect on consumption. Children in these age brackets are dependents and are expected to attend school. Thus, they comprise a significant portion of household consumption. Families with many dependent children are expected to be needing more resources. The variable *Elementary* is also positively significant, implying the low educational level of beneficiary household heads. Education is an indicator of potential command over resources. Less educated household heads have lower chances of being employed in high-paying jobs. They may resort to engaging in business but with low returns. *Spousewrkdum* is positive and significantly related to program participation. The insufficiency of income among household beneficiaries requires the spouse to work to support the needs of the family.

The housing characteristics such as *Rooflight*, *Walllight*, and *Singhousedum* also positively and significantly predict participation in the CCT program. Poor housing condition indicates the incapability of households to spend on high-quality housing materials and make significant renovations to their houses. This condition increases the vulnerability of the households to disasters, increasing their chances of being poor. In addition, beneficiaries find it difficult to access basic utilities such as water, as indicated by a significant positive sign for *Water distance*. On the other hand, other housing characteristics, such as *Notoilet* and *Ownhouselot*, are not significant. Toilet facilities are already accessible to both beneficiaries and non-beneficiaries. Owning a house and lot among beneficiaries is also not a problem. This is probably due to the fact that most beneficiaries are located in rural areas where land ownership is not a major concern (Tutor, 2014).

The variable *OFW* significantly and negatively predicts membership in the program. Cash from household members abroad is a significant augmentation in household income. The appliance also negatively predicts program participation. Obviously, non-beneficiary who are considered as non-poor have more capacity to buy home appliances.

Overall, the characteristics of household targeted by the CCT program is affirmed by the estimation of the propensity score.

Balance Test

The balancing condition must be established when using the propensity score to simulate the randomization effect and satisfy the CIA and matching assumption. This allows researchers to determine which among the covariates are not balanced. As previously mentioned, a perfectly balanced covariate has a standardized difference of zero and a variance ratio of one. Therefore, variables that are nowhere near these values are marked and subjected to transformation or re-categorization if a further test reveals that, indeed, the covariates are unbalanced. Moreover, re-specification of the propensity model can also be done if needed to ensure the balance of the covariates.

Table 5 presents the standardized difference and the variance ratio for each covariate.

Table 5. Standardized difference and the variance ratio for each covariate

Covariates	Standardized Differences		Variance Ratio	
	Raw	Weighted	Raw	Weighted
Male	0.208	-0.005	0.660	1.012

	Standardized Differences		Variance Ratio	
Covariates	Raw	Weighted	Raw	Weighted
Married	0.283	0.000	0.605	0.999
Age	-0.429	-0.004	0.536	0.991
Age2	-0.474	-0.005	0.484	1.037
Elementary	0.127	-0.020	0.949	1.011
Hmembers0_5dum	0.168	0.002	1.014	1.000
Hmembers6_14dum	0.961	-0.006	0.244	1.022
Hmembers15_18dum	0.399	-0.043	1.187	1.001
Hhwork	0.314	0.000	0.430	1.002
Spousewrkdum	0.247	-0.013	1.027	1.002
Floor_area	-0.099	0.043	0.723	1.090
Rooflight	0.238	-0.035	1.429	0.963
Walllight	0.271	-0.032	1.253	0.984
Water_distance	0.138	-0.019	1.218	0.740
Notoilet	0.043	-0.016	1.179	0.945
Ownhouselot	0.005	-0.011	0.998	1.005
Renthlot	-0.187	0.005	0.304	1.045
Singhousedum	0.167	-0.004	0.305	1.036
OFW	-0.110	-0.012	0.702	0.958
NCR	-0.132	-0.005	0.438	0.965
Appliance	-0.203	0.029	0.670	0.847

Results show that the weighted standardized difference of all the covariates is near the value of zero. However, the variance ratio shows that the variables. *Water_distance* and *Appliance* have values 0.740 and 0.847, which are considered not close to one. This indicates that either one between non-beneficiary households and beneficiary households has more dispersed values in *Water_distance* and *Appliance*.

However, this balance check is insufficient to warrant the transformation of the variables and re-specification of the propensity model. A more formal test proposed by Imai and Ratkovic (2014) is implemented to determine whether the covariates are indeed statistically unbalanced. The null hypothesis that the covariates are balanced is tested. Results indicate a value of 31.9113 with a p-value of 0.0789. Since the p-value is greater than 0.05, the null hypothesis that the covariates are balanced is not rejected. It implies that the balancing condition holds, and the estimation of the causal effect of the CCT program is permissible.

Empirical Results

The effect of CCT on savings, borrowings, and participation in enterprises is estimated using the parameters POM, ATE, and ATET and the methods IPW, AIPW, IPWRA, and NNM whenever applicable.

The coefficient is interpreted differently for each outcome since the study uses several measures for each outcome. Table 6 presents the summary of the interpretation of the coefficient.

Table 6. Summary of the interpretation of the coefficient by type of outcomes

Type of Outcomes	Interpretation of:		
	POM	ATE	ATET
Binary outcomes (i.e., <i>Savingsdum</i> , <i>Creditdum</i> , <i>Entrep</i>)	Proportion or the probability of the household in the treatment indicating the outcome	The average change in the probability of the household manifests the outcome between the beneficiary and non-beneficiary households	Change in the probability of the beneficiary household to manifest the outcome had they not participated in the CCT program.
Count and continuous outcomes (i.e., <i>Savings_rate</i> , <i>Num_credit</i> , <i>Num_entrep</i>)	The average value of the outcome for each treatment.	The average change in the outcome between the beneficiary and non-beneficiary households	The average change in the beneficiary households' outcome had they not participated in the CCT program.

For binary outcomes (*Entrep*), the coefficient of the POM is the proportion or the probability of the

household in the treatment indicating the outcome. The coefficient of the ATE is the causal effect. It is the difference in the POMs of the beneficiary and non-beneficiary households. It is interpreted as the average change in the probability of the household manifesting the outcome between the beneficiary and non-beneficiary households. The coefficient of the ATET is the change in the probability of the beneficiary household manifesting the outcome had they not participated in the CCT program. For continuous outcome and count outcomes (*Num_entrep*), the POM is the average value of the outcome for each treatment. The coefficient of the ATE is the average change in the outcome between the beneficiary and non-beneficiary households. The coefficient of the ATET is the average change in the outcome of the beneficiary households had they not participated in the CCT program.

Impact on Participation in Enterprises

Table 7 presents the estimate of the causal effect of CCT on the outcome enterprise participation.

Table 7. Estimate of the causal effect of CCT on enterprise participation

Parameter	Estimators							
	IPW		AIPW		IPWRA		NNM	
	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.	Coef.	S.E.
Entrep								
POM								
Non-Beneficiary	0.594***	0.010	0.593***	0.010	0.593***	0.010	n/a	
Beneficiary	0.660***	0.017	0.663***	0.015	0.666***	0.014	n/a	
ATE	0.066***	0.019	0.070***	0.018	0.073***	0.017	0.054***	0.016
ATET	0.084***	0.018	n/a		0.085***	0.017	0.091***	0.018
Num_entrep								
POM								
Non-Beneficiary	0.764***	0.016	0.762***	0.016	0.762***	0.016	n/a	
Beneficiary	0.906***	0.032	0.914***	0.028	0.916***	0.026	n/a	
ATE	0.143***	0.035	0.153***	0.031	0.154***	0.030	0.098***	0.027
ATET	0.147***	0.028	n/a		0.149***	0.028	0.138***	0.031

* p<0.1; ** p<0.05; *** p<0.01; ns not significant

The parameters of the causal effect of CCT on the *Entrep* generally reveal a positive impact. The POM shows that beneficiary households have higher chances of participating in any enterprise than non-beneficiary households. Beneficiary households have a 66% to 66.6% chance of participating in any enterprise. Meanwhile, non-beneficiary households have only a 59.3% to 59.4% chance of participating in any enterprise. This difference in the POM is highly significant, with the ATE ranging from 0.054 to 0.073. This result implies that beneficiary households have a higher chance of participating in any enterprise by 5.4% to 7.3% than non-beneficiary households. The ATET shows that the chances of enterprise participation are higher by 8.4% to 9.1% if households participate in the CCT program than if they had not participated in the program.

In terms of *Num_entrep* or the number of enterprises participated by the household, a positive impact is also observed. Beneficiary households have an average number of participation in enterprises of 0.906 to 0.916 times. Meanwhile, non-beneficiary households participate in enterprise on an average of 0.762 to 0.764 times. The difference in this average number of participation in enterprises is also significant. The ATE is 0.143 to 0.154. This result means that beneficiary households participate in an enterprise by 0.098 to 0.154 times more than non-beneficiary households. Moreover, the ATET also shows that participation of beneficiary households in enterprises is more by 0.138 to 0.149 times than if they had not participated in the CCT program.

The result of the analysis supports the proposition of the study, which states that cash transfers have a positive impact on enterprise participation. This result supports earlier empirical studies (Gertler et al., 2012; Veras Soares et al., 2010) and provides local evidence contradicting the findings of Chaudhury, Friedman, and Onishi (2013). As stated by Barrientos (2012), CCT secures consumption and asset. Income primarily devoted to consumption before the program can now be saved and invested. Households' confidence to engage in risky activities increases because of the cash grants. Beneficiary households may also be encouraged to enter markets because of the local community effects of CCT. Usually, beneficiary households are concentrated in a particular area, and the additional income will increase local community demand for goods and services that will encourage entry.

Conclusion

This study seeks to determine if CCT has a positive impact on entrepreneurial behavior. The country implemented its own CCT program in 2008, and it has become the flagship strategy against poverty. Poor households do not have access to credit and are illiquid, limiting their consumption and investment. Steady streams of cash from the program alleviate credit and liquidity constraints allowing them to borrow, save, and allocate resources for more productive purposes (Barrientos, 2012). Therefore, CCTs are hypothesized to positively affect participation in enterprises. Empirical studies suggest that CCT's impact on participation in business is still inconclusive.

Results show that CCT has a positive effect on participation in enterprises. CCT increases both the probability of households engaging in any form of enterprise and the number of enterprises they participate. Cash transfers increase the confidence of poor households to engage in business leading to more efficient allocation of resources.

This study has shown that CCT has an impact on investment in business among households. These behaviors have important long-term implications on the households' living standards. Participation and success in business increase the chance of households becoming productive after graduating from the program. In the long run, these behaviors can improve the living standard of the household. Hence, policymakers can integrate a comprehensive mechanism into the CCT program to promote entrepreneurial capabilities.

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